Exercise 4.3

Elena Song

2026-11-07

Question 1

As per Exercise 4.2, for X = 1, 2, ..., 100, simulate Y = 2 + 0.5X + epsilon, where epsilon ~ N(0, sigma = 15). Set the seed to 1.

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.2.0     v readr     2.2.0
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.2     v tibble    3.3.1
## v lubridate 1.9.5     v tidyr     1.3.2
## v purrr     1.2.1
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
# Simulate data
set.seed(1)

data1 <- data.frame(X = seq(1, 100, 1)) %>%
  mutate(
    EY = 2 + 0.5 * X,
    e = rnorm(100, sd = 15),
    Y = EY + e
  )
```

1. Fit a simple linear regression model. Save the raw residuals.

```
fit1 <- lm(Y ~ X, data = data1)
summary(fit1)
```

```
##
## Call:
## lm(formula = Y ~ X, data = data1)
##
## Residuals:
```

```
##     Min      1Q  Median      3Q     Max
## -35.101  -9.088   0.233   8.777  34.462
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  3.97499    2.72846   1.457    0.148
## X            0.49323    0.04691  10.515   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 13.54 on 98 degrees of freedom
## Multiple R-squared:  0.5301, Adjusted R-squared:  0.5253
## F-statistic: 110.6 on 1 and 98 DF,  p-value: < 2.2e-16
```

```
raw_resid <- resid(fit1)
```

Answer: I fitted a simple linear regression model of Y on X using lm() and saved the raw residuals using resid().

2. Based on the regression coefficients, is the fitted model similar to the simulated specifications?

```
coef(fit1)
```

```
## (Intercept)           X
##   3.9749860   0.4932341
```

Answer: Yes, the fitted model is reasonably similar to the simulated specifications. The estimated slope(0.493) is very close to true slope (0.5), while estimated intercept(3.975) differs somewhat from the true intercept (2). These differences are expected due to random variation in the simulated data.

3. For the error term , find the mean, the standard deviation, and create a Q-Q plot. Do the same for the residuals. You should find that the results are similar but not identical. Why?

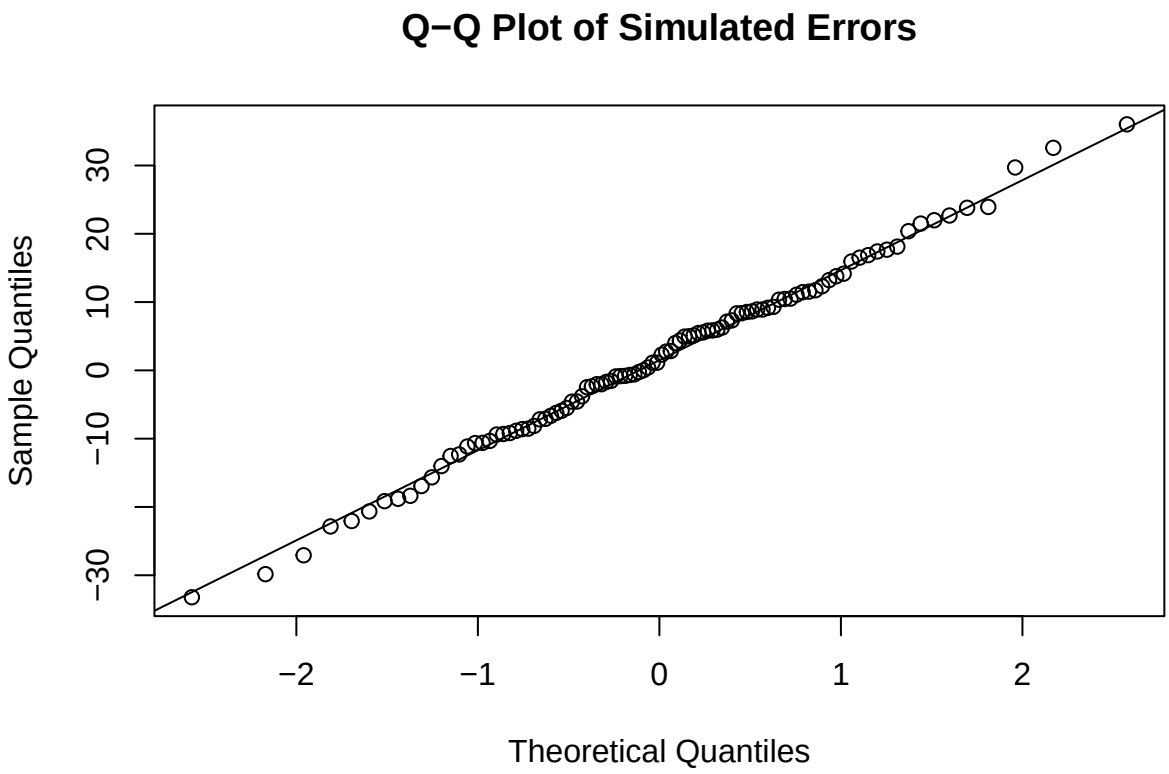
```
# Simulated errors
mean(data1$e)
```

```
## [1] 1.633311
```

```
sd(data1$e)
```

```
## [1] 13.47299
```

```
qqnorm(data1$e, main = "Q-Q Plot of Simulated Errors")
qqline(data1$e)
```



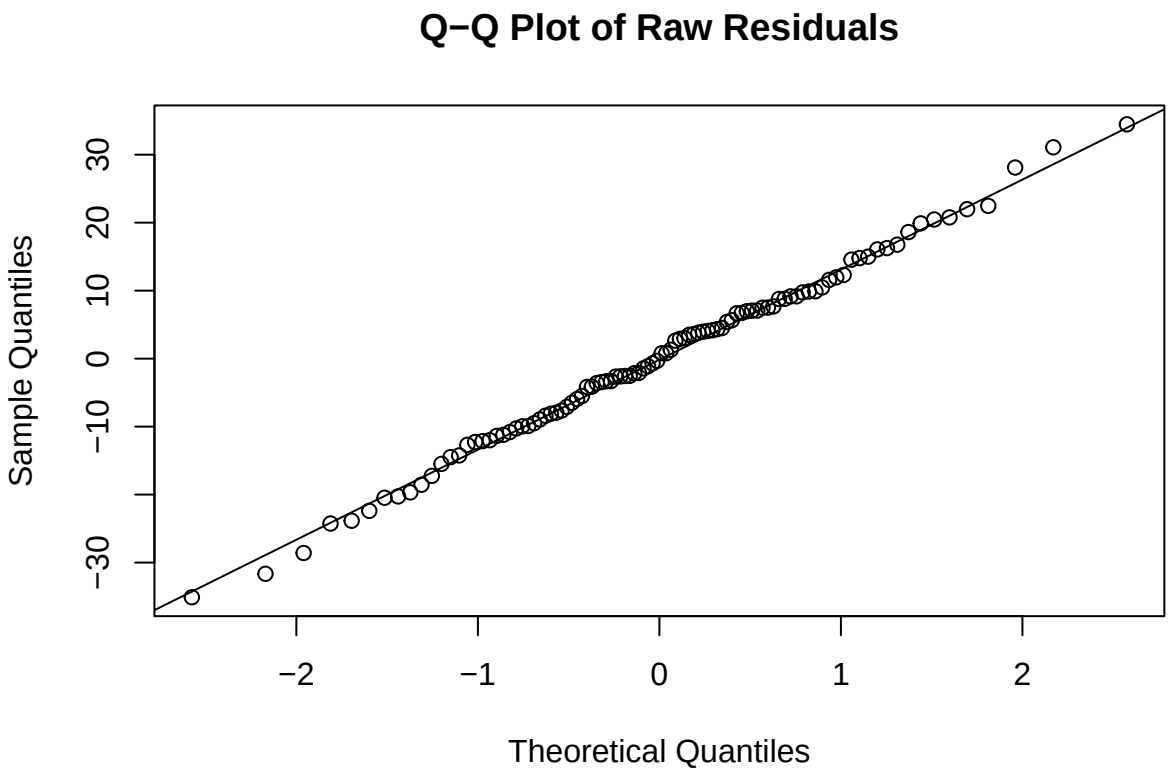

```r
# Raw residuals
mean(raw_resid)
```

```
## [1] 4.498311e-16
```

```r
sd(raw_resid)
```

```
## [1] 13.47156
```

```r
qqnorm(raw_resid, main = "Q-Q Plot of Raw Residuals")
qqline(raw_resid)
```



Answer The simulated errors had a mean of 1.6333 and a standard deviation of 13.4730, while the raw residuals had a mean of approximately 0 and a standard deviation of 13.4716. Both Q-Q plots show points closely following the reference line, suggesting that the errors and residuals are approximately normally distributed. The results are similar but not identical because the residuals are calculated using the fitted regression coefficients rather than the true coefficients. In addition, the residuals have a mean of approximately zero because OLS model includes an intercept.

4. From what we know about normal distributions, how many observations should have values of that exceed two standard deviations away from the mean? How many do you see?

```
# Expected number beyond 2 standard deviations
100*2*pnorm(-2)
```

```
## [1] 4.550026
```

```
# Observed number using the true population SD (15)
sum(abs(data1$e -0) >2 *15)
```

```
## [1] 3
```

```
# Observed number using the sample mean and SD
sum(abs(data1$e - mean(data1$e)) > 2 * sd(data1$e))
```

```
## [1] 6
```

Answer For a normal distribution, approximately 4.55% of observations are expected to fall more than two standard deviations from the mean, corresponding to about 4 to 5 out of 100 observations. In the simulated data, 3 observations exceeded two population standard deviations from the mean (±30), while 6 exceeded two sample standard deviations from the sample mean. The difference arises because the sample mean and standard deviation differ from the population parameters.

5. Do the same for the residuals. Depending on the cut-point you use, these observations would be classified as "potential outliers". However, because we know the specifications of the simulation, we also know that these data points are legitimate.

```r
#calculate mean and SD of raw residuals
mean(raw_resid)
```

```
## [1] 4.498311e-16
```

```r
sd(raw_resid)
```

```
## [1] 13.47156
```

```r
#count residuals beyond 2 standard deviations
sum(abs(raw_resid - mean(raw_resid)) > 2 * sd(raw_resid))
```

```
## [1] 6
```

Answer The raw residuals had a mean of approximately 0 and a standard deviation of 13.472. Six observations had residuals more than two standard deviations away from the mean. Although these observations could be classified as potential outliers, they were generated from the specified simulation model and are therefore legitimate observations. This shows that extreme residuals can occur naturally, even when the model assumptions are satisfied.

6. Calculate the studentized residuals, and sort this dataset by their absolute values. Which observations have studentized residuals that exceed 2 in absolute value?

```r
# calculate studentized residuals
resid_stu <- rstudent(fit1)

# sort by absolute studentized residuals
sort(abs(resid_stu), decreasing = TRUE) %>% head(10)
```

```
##       14       61       24       70       67       56       84       28
## 2.710428 2.636270 2.417148 2.366546 2.165764 2.123942 1.834890 1.795561
##       95       35
## 1.704617 1.680072
```

```r
# Identify observations with |studentized residual| > 2
which(abs(resid_stu) > 2)
```

```
## 14 24 56 61 67 70
## 14 24 56 61 67 70
```

Answer After calculating and sorting the absolute studentized residuals, six observations exceeded 2 in absolute value: observations 14,24,56,61,67, and 70. These observations may be considered potential outliers, although they were generated from the specified simulation model.

7. Print the leverage statistics. Which points have the greatest leverage? Do they have notably different leverage than the other data points?

```r
# Calculate leverage statistics
leverage1 <- hatvalues(fit1)
leverage1
```

```
##          1          2          3          4          5          6          7
## 0.03940594 0.03822982 0.03707771 0.03594959 0.03484548 0.03376538 0.03270927
##          8          9         10         11         12         13         14
## 0.03167717 0.03066907 0.02968497 0.02872487 0.02778878 0.02687669 0.02598860
##         15         16         17         18         19         20         21
## 0.02512451 0.02428443 0.02346835 0.02267627 0.02190819 0.02116412 0.02044404
##         22         23         24         25         26         27         28
## 0.01974797 0.01907591 0.01842784 0.01780378 0.01720372 0.01662766 0.01607561
##         29         30         31         32         33         34         35
## 0.01554755 0.01504350 0.01456346 0.01410741 0.01367537 0.01326733 0.01288329
##         36         37         38         39         40         41         42
## 0.01252325 0.01218722 0.01187519 0.01158716 0.01132313 0.01108311 0.01086709
##         43         44         45         46         47         48         49
## 0.01067507 0.01050705 0.01036304 0.01024302 0.01014701 0.01007501 0.01002700
##         50         51         52         53         54         55         56
## 0.01000300 0.01000300 0.01002700 0.01007501 0.01014701 0.01024302 0.01036304
##         57         58         59         60         61         62         63
## 0.01050705 0.01067507 0.01086709 0.01108311 0.01132313 0.01158716 0.01187519
##         64         65         66         67         68         69         70
## 0.01218722 0.01252325 0.01288329 0.01326733 0.01367537 0.01410741 0.01456346
##         71         72         73         74         75         76         77
## 0.01504350 0.01554755 0.01607561 0.01662766 0.01720372 0.01780378 0.01842784
##         78         79         80         81         82         83         84
## 0.01907591 0.01974797 0.02044404 0.02116412 0.02190819 0.02267627 0.02346835
##         85         86         87         88         89         90         91
## 0.02428443 0.02512451 0.02598860 0.02687669 0.02778878 0.02872487 0.02968497
##         92         93         94         95         96         97         98
## 0.03066907 0.03167717 0.03270927 0.03376538 0.03484548 0.03594959 0.03707771
##         99        100
## 0.03822982 0.03940594
```

```r
sort(leverage1, decreasing = TRUE) %>% head(10)
```

```
##           1         100           2          99           3          98           4
## 0.03940594 0.03940594 0.03822982 0.03822982 0.03707771 0.03707771 0.03594959
##          97           5          96
## 0.03594959 0.03484548 0.03484548
```

```r
summary(leverage1)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.01000 0.01188 0.01750 0.02000 0.02688 0.03941
```

Answer Observations 1 (X = 1) and 100 (X = 100) have the highest leverage values, both approximately 0.0394. These observations are farthest from the mean of X (50.5). However, their leverage values are not notably different from those of nearby observations, and no point has unusually high leverage.

Question 2

In addition to the previous observations from Question 1, add another ten observations where X = 5, 15, ..., 95, Y = 2 + 0.5X + epsilon and epsilon ~ N(0, sigma = 150). Many, albeit not all, of these 10 observations will appear to be outliers. Again, set the seed to 1.

```r
# Add a group indicator to the original 100 observations
data1$question <- 1

# Set the seed and advance past the original 100 errors
set.seed(1)
invisible(rnorm(100, sd = 15))

# Generate 10 additional observations
data_extra <- data.frame(X = seq(5, 95, 10)) %>%
  mutate(
    EY = 2 + 0.5 * X,
    e = rnorm(10, sd = 150),
    Y = EY + e,
    question = 2
  )

# Combine datasets
data2 <- rbind(data1, data_extra)

nrow(data2)
```

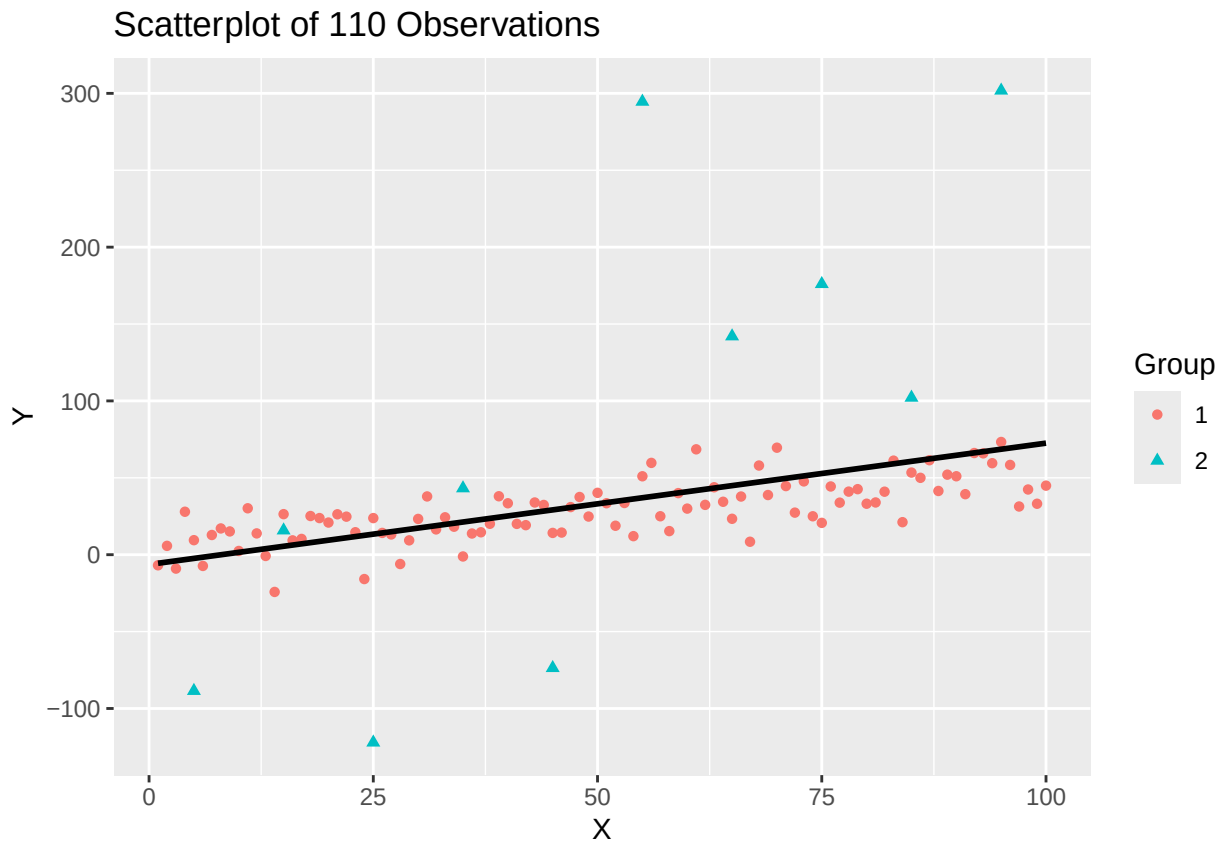
```
## [1] 110
```

1. Create a scatterplot for the 110 data points, and append the best-fitting regression line. Using different plotting characters for the 2 subpopulations.

```r
ggplot(data = data2, aes(x = X, y = Y)) +
  geom_point(aes(color = factor(question),
                 shape = factor(question))) +
  geom_smooth(method = "lm", se =FALSE, color = "black") +
```

```
  labs(
    title = "Scatterplot of 110 Observations",
    x = "X",
    y = "Y",
    color = "Group", shape = "Group"
  )
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



Answer The scatterplot shows that the additional 10 observations (Group 2) are generally more widely scattered around the fitted regression line than the original 100 observations (Group 1). Several of these observations appear to be outliers, although some remain close to the line. This is consistent with the larger error standard deviation used to generate Group 2.

2. Fit a simple linear model. In comparison to the previous regression, what has happened to the regression coefficients?

```
# Fit regression model using all 110 observations
fit2 <-  lm(Y ~ X, data = data2)
summary(fit2)
```

```
##
## Call:
```

```
## lm(formula = Y ~ X, data = data2)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -135.510  -14.775   -2.976   10.402  257.557
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -6.3480     8.4969  -0.747    0.457
## X             0.7888     0.1462   5.396 4.07e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 44.24 on 108 degrees of freedom
## Multiple R-squared:  0.2123, Adjusted R-squared:  0.205
## F-statistic: 29.11 on 1 and 108 DF,  p-value: 4.071e-07
```

```
#compare regression coefficients
coef(fit1)
```

```
## (Intercept)           X
##   3.9749860   0.4932341
```

```
coef(fit2)
```

```
## (Intercept)           X
##  -6.3479649   0.7887977
```

Answer After adding the 10 observations with larger error variability, the estimated intercept decreased from 3.975 to -6.348, while the estimated slope increased from 0.493 to 0.789. The additional observations changed the fitted regression line, making it steeper and moving the slope estimate farther from the true value of 0.5.

3. In comparison to the previous regression, what has happened to the R-squared?

```
# Compare R-squared values
summary(fit1)$r.squared
```

```
## [1] 0.5301334
```

```
summary(fit2)$r.squared
```

```
## [1] 0.2123316
```

Answer The R-squared decreased from 0.5301 in Question 1 to 0.2123 in Question 2. This indicates that the proportion of variation in Y explained by X decreased from approximately 53.0% to 21.2%. The additional observations introduced much greater error variability, reducing the overall fit of the regression model.

4. Which observations have studentized residuals that exceed 2 in absolute value?

```r
# Calculate studentized residuals
resid_stu2 <- rstudent(fit2)

# Sort by absolute studentized residuals
sort(abs(resid_stu2), decreasing = TRUE) %>% head(10)
```

```
##       106       110       103       108       105       107       101       109
## 7.0436140 6.2196635 3.2192442 2.9020929 2.3857227 2.2481207 2.0064722 0.9465203
##        97        99
## 0.8912429 0.8875285
```

```r
# Identify observations with |studentized residual| > 2
which(abs(resid_stu2) > 2)
```

```
## 101 103 105 106 107 108 110
## 101 103 105 106 107 108 110
```

```r
# Display the observations and their groups
data2 %>%
  mutate(studentized_resid = resid_stu2) %>%
  filter(abs(studentized_resid) > 2) %>%
  select(X, Y, question, studentized_resid)
```

```
##    X         Y question studentized_resid
## 1  5  -88.5550        2         -2.006472
## 2 25 -122.1382        2         -3.219244
## 3 45  -73.6877        2         -2.385723
## 4 55  294.5931        2          7.043614
## 5 65  142.0061        2          2.248121
## 6 75  176.0261        2          2.902093
## 7 95  301.8264        2          6.219664
```

Answer Seven observations had absolute studentized residuals greater than 2: observations 101, 103, 105, 106, 107, 108, and 110. All seven belong to the additional group with larger error variability, indicating that these observations are potential outliers.

5. Are there any high leverage values for this set of observations?

```r
# Calculate leverage statistics
leverage2 <- hatvalues(fit2)
leverage2
```

```
##           1           2           3           4           5           6
## 0.035797886 0.034728744 0.033681441 0.032655979 0.031652355 0.030670571
##           7           8           9          10          11          12
## 0.029710627 0.028772522 0.027856257 0.026961831 0.026089244 0.025238497
##          13          14          15          16          17          18
## 0.024409590 0.023602521 0.022817293 0.022053904 0.021312354 0.020592644
```

```
##          19          20          21          22          23          24
## 0.019894773 0.019218742 0.018564551 0.017932198 0.017321686 0.016733012
##          25          26          27          28          29          30
## 0.016166179 0.015621184 0.015098029 0.014596714 0.014117238 0.013659602
##          31          32          33          34          35          36
## 0.013223805 0.012809848 0.012417730 0.012047451 0.011699012 0.011372413
##          37          38          39          40          41          42
## 0.011067653 0.010784732 0.010523651 0.010284410 0.010067007 0.009871445
##          43          44          45          46          47          48
## 0.009697722 0.009545838 0.009415794 0.009307589 0.009221224 0.009156698
##          49          50          51          52          53          54
## 0.009114012 0.009093165 0.009094158 0.009116990 0.009161662 0.009228173
##          55          56          57          58          59          60
## 0.009316524 0.009426714 0.009558743 0.009712612 0.009888321 0.010085869
##          61          62          63          64          65          66
## 0.010305256 0.010546483 0.010809550 0.011094456 0.011401201 0.011729786
##          67          68          69          70          71          72
## 0.012080210 0.012452474 0.012846578 0.013262520 0.013700303 0.014159925
##          73          74          75          76          77          78
## 0.014641386 0.015144687 0.015669827 0.016216806 0.016785626 0.017376284
##          79          80          81          82          83          84
## 0.017988782 0.018623120 0.019279297 0.019957314 0.020657170 0.021378865
##          85          86          87          88          89          90
## 0.022122400 0.022887775 0.023674989 0.024484042 0.025314935 0.026167668
##          91          92          93          94          95          96
## 0.027042240 0.027938651 0.028856902 0.029796992 0.030758922 0.031742691
##          97          98          99         100         101         102
## 0.032748300 0.033775748 0.034825036 0.035896163 0.031652355 0.022817293
##         103         104         105         106         107         108
## 0.016166179 0.011699012 0.009415794 0.009316524 0.011401201 0.015669827
##         109         110
## 0.022122400 0.030758922
```

```
# Sort leverage values in descending order
sort(leverage2, decreasing = TRUE) %>% head(10)
```

```
##        100          1         99          2         98          3         97
## 0.03589616 0.03579789 0.03482504 0.03472874 0.03377575 0.03368144 0.03274830
##          4         96          5
## 0.03265598 0.03174269 0.03165236
```

```
summary(leverage2)
```

```
##     Min.  1st Qu.   Median     Mean  3rd Qu.     Max.
## 0.009093 0.010874 0.015918 0.018182 0.024226 0.035896
```

```
# Common screening threshold: 2p/n
2 * 2 / nrow(data2)
```

```
## [1] 0.03636364
```

Answer There are no unusually high leverage values in this dataset. The highest leverage occurs at observations 100 and 1, with values of approximately 0.0359 and 0.0358, respectively. Both values are below the

common screening threshold of 0.0364 (2p/n). Although several additional observations have large studentized residuals, their X values remain within the original range of 1 to 100, so they do not have unusually high leverage.

Question 3

Add one more observation to the original 100, having X = 500, Y = 252.

```
# Original 100 observations from Question 1
data3 <- data1 %>%
  select(X, Y)

# Add the new observation
new_point <- data.frame(X = 500, Y = 252)

data3 <- rbind(data3, new_point)

# Check the number of observations
nrow(data3)
```

```
## [1] 101
```

1. Fit a regression model. What are the regression coefficients?

```
# Fit regression model with 101 observations
fit3 <- lm(Y ~ X, data = data3)
summary(fit3)
```

```
##
## Call:
## lm(formula = Y ~ X, data = data3)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -35.024  -8.908   0.410   8.766  34.435
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  3.86742    1.93153   2.002    0.048 *
## X            0.49545    0.02531  19.577   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 13.47 on 99 degrees of freedom
## Multiple R-squared:  0.7947, Adjusted R-squared:  0.7926
## F-statistic: 383.3 on 1 and 99 DF,  p-value: < 2.2e-16
```

```
# Compare coefficients with Question 1
coef(fit1)
```

```
## (Intercept)           X
##   3.9749860   0.4932341
```

```
coef(fit3)
```

```
## (Intercept)           X
##   3.8674186   0.4954454
```

Answer The fitted regression model has an intercept of 3.8674 and a slope of 0.4954. Compared with the original model (intercept = 3.9750, slope = 0.4932), the regression coefficients changed very little after adding the observation (500, 252).

2. What is the R-squared?

```
# Compare R-squared values
summary(fit1)$r.squared
```

```
## [1] 0.5301334
```

```
summary(fit3)$r.squared
```

```
## [1] 0.7947201
```

Answer The R-squared is 0.7947, compared with 0.5301 in the original model. Although the R-squared increased substantially after adding the new observation, the regression coefficients changed very little.

3. What is the residual for the point with X = 500?

```
# Calculate the residual for X = 500
resid(fit3)[101]
```

```
##       101
## 0.4098989
```

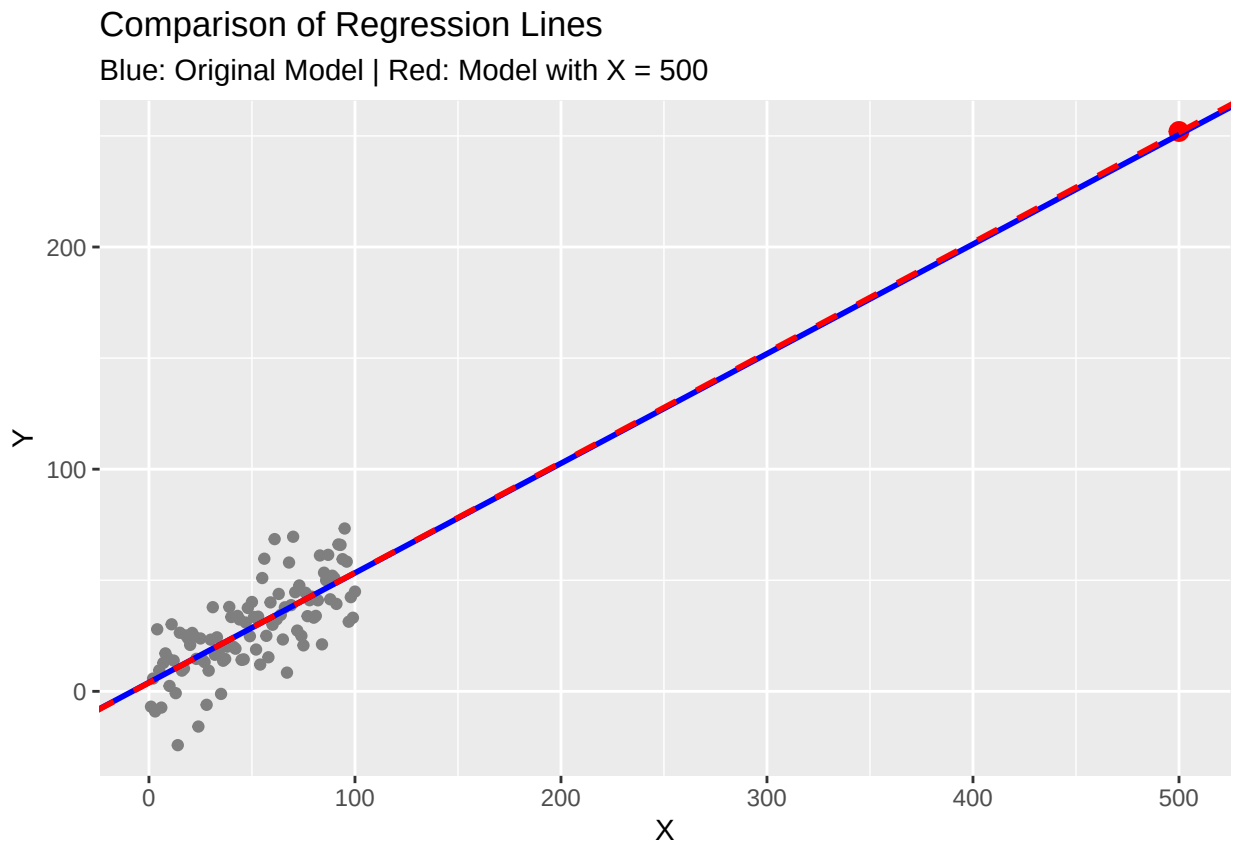
```
# Verify using observed minus predicted Y
data3$Y[101] - predict(fit3, newdata = data3[101, ])
```

```
##       101
## 0.4098989
```

Answer The residual for the observation with X = 500 is approximately 0.4099. This indicates that the observed Y value (252) is very close to the fitted value (251.5901), despite the observation having an extreme X value.

4. Create a scatterplot with the original 100 data points, the original regression line (from Question 1), and the new regression line (from Question 3). How similar are they?

```r
ggplot(data = data1, aes(x = X, y = Y)) +
  geom_point(color = "gray50") +
  geom_point(data = data3[101, ],
             aes(x = X, y = Y),
             color = "red", size = 3) +
  geom_abline(intercept = coef(fit1)[1],
              slope = coef(fit1)[2],
              color = "blue", linewidth = 1) +
  geom_abline(intercept = coef(fit3)[1],
              slope = coef(fit3)[2],
              color = "red", linetype = "dashed",
              linewidth = 1) +
  labs(
    title = "Comparison of Regression Lines",
    subtitle = "Blue: Original Model | Red: Model with X = 500",
    x = "X",
    y = "Y"
  )
```



Answer The two regression lines are nearly identical and almost completely overlap in the scatterplot. Although the new observation has an extreme X value of 500, it lies close to the original regression line and causes only minimal changes in the estimated intercept and slope.

5. What is the leverage value of the X = 500 point? Is it much different from the other leverage values?

```
# Calculate leverage statistics
leverage3 <- hatvalues(fit3)
leverage3[101]
```

```
##       101
## 0.7088661
```

```
# Compare with other observations
sort(leverage3, decreasing = TRUE) %>% head(10)
```

```
##        101          1          2          3          4          5          6
## 0.70886609 0.02017239 0.01979515 0.01942496 0.01906184 0.01870577 0.01835676
##          7          8          9
## 0.01801480 0.01767991 0.01735207
```

```
# Summary of leverage values
summary(leverage3)
```

```
##     Min.  1st Qu.   Median     Mean  3rd Qu.     Max.
## 0.009901 0.010493 0.012115 0.019802 0.014983 0.708866
```

Answer The leverage value of the observation with X = 500 is approximately 0.7089, which is substantially higher than the leverage values of the other observations. The next highest leverage is approximately 0.0202. This indicates that the new observation has extremely high leverage because its X value is far outside the range of the original data.

6. Here, X = 500 is a high leverage point but not an influential point. Why?

Answer The observation with X = 500 has high leverage because its X value is far from the other observations. However, it is not influential in terms of the regression coefficients because it lies very close to the original regression line. Its residual is only 0.4099, and adding this observation causes minimal changes in the estimated intercept and slope.

Question 4

Now, replace the point (X = 500, Y = 252) with (X = 500, Y = -600).

```
# Copy the Question 3 dataset
data4 <- data3

# Replace Y = 252 with Y = -600 for X = 500
data4$Y[data4$X == 500] <- -600

# Check the updated observation
tail(data4)
```

```
##       X          Y
## 96   96   58.37730
## 97   97   31.35112
## 98   98   42.40102
## 99   99   33.13081
## 100 100   44.89899
## 101 500 -600.00000
```

```r
# Confirm the number of observations
nrow(data4)
```

```
## [1] 101
```

1. Fit a regression model. What are the regression coefficients?

```r
# Fit regression model with the modified observation
fit4 <- lm(Y ~ X, data = data4)
summary(fit4)
```

```
##
## Call:
## lm(formula = Y ~ X, data = data4)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -247.636  -29.515    1.228   37.782   84.393
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 68.96073    6.88971  10.009  < 2e-16 ***
## X           -0.84265    0.09027  -9.335 3.09e-15 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 48.05 on 99 degrees of freedom
## Multiple R-squared:  0.4681, Adjusted R-squared:  0.4628
## F-statistic: 87.14 on 1 and 99 DF,  p-value: 3.092e-15
```

```r
# Compare regression coefficients
coef(fit1)
```

```
## (Intercept)           X
##   3.9749860   0.4932341
```

```r
coef(fit3)
```

```
## (Intercept)           X
##   3.8674186   0.4954454
```

```r
coef(fit4)
```

```
## (Intercept)           X
##  68.9607264  -0.8426491
```

Answer The fitted regression model has an intercept of 68.9607 and a slope of −0.8426. Compared with the original model (intercept = 3.9750, slope = 0.4932), the regression coefficients changed substantially. In particular, the slope changed from positive to negative after replacing the observation with (500, −600).

2. What is the R-squared?

```r
# Compare R-squared values
summary(fit1)$r.squared
```

```
## [1] 0.5301334
```

```r
summary(fit3)$r.squared
```

```
## [1] 0.7947201
```

```r
summary(fit4)$r.squared
```

```
## [1] 0.4681354
```

Answer The R-squared is 0.4681, indicating that approximately 46.8% of the variation in Y is explained by the fitted regression model. This is lower than the R-squared of 0.7947 in Question 3 and 0.5301 in the original model.

3. What is the residual for the point with X = 500?

```r
# Calculate the residual for X = 500
resid(fit4)[101]
```

```
##       101
## -247.6362
```

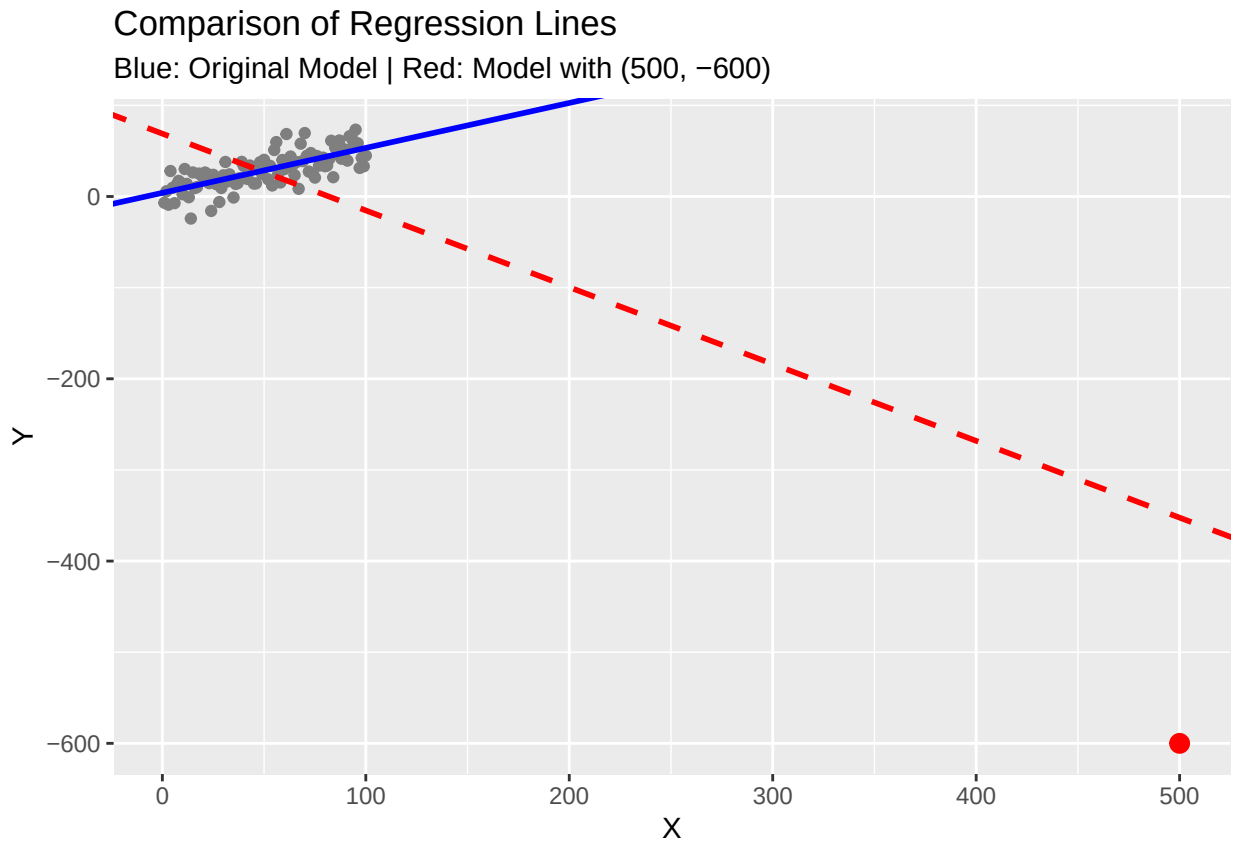
```r
# Verify using observed minus predicted Y
data4$Y[101] - predict(fit4, newdata = data4[101, ])
```

```
##       101
## -247.6362
```

Answer The residual for the observation with X = 500 is approximately −247.6362. The observed Y value (−600) is substantially lower than the fitted value (approximately −352.3638). This indicates that the observation lies far below the fitted regression line.

4. Create a scatterplot with the original 100 data points, the original regression line (from Question 1), and the new regression line (from Question 4). How similar are they?

```
ggplot(data = data1, aes(x = X, y = Y)) +
  geom_point(color = "gray50") +
  geom_point(data = data4[101, ],
             aes(x = X, y = Y),
             color = "red", size = 3) +
  geom_abline(intercept = coef(fit1)[1],
              slope = coef(fit1)[2],
              color = "blue", linewidth = 1) +
  geom_abline(intercept = coef(fit4)[1],
              slope = coef(fit4)[2],
              color = "red", linetype = "dashed",
              linewidth = 1) +
  labs(
    title = "Comparison of Regression Lines",
    subtitle = "Blue: Original Model | Red: Model with (500, -600)",
    x = "X",
    y = "Y"
  )
```



Answer The two regression lines are substantially different. The original regression line has a positive slope of 0.4932, while the new regression line has a negative slope of −0.8426. Adding the observation (500, −600) reverses the direction of the fitted relationship, indicating that this observation is highly influential.

5. What is the leverage value of the X = 500 point? Is it much different from the other leverage values?

```
# Calculate leverage statistics
leverage4 <- hatvalues(fit4)
leverage4[101]
```

```
##       101
## 0.7088661
```

```
# Compare with other observations
sort(leverage4, decreasing = TRUE) %>% head(10)
```

```
##        101          1          2          3          4          5          6
## 0.70886609 0.02017239 0.01979515 0.01942496 0.01906184 0.01870577 0.01835676
##          7          8          9
## 0.01801480 0.01767991 0.01735207
```

```
# Compare leverage with Question 3
leverage3[101]
```

```
##       101
## 0.7088661
```

```
leverage4[101]
```

```
##       101
## 0.7088661
```

Answer The leverage value of the observation with X = 500 is approximately 0.7089, which is substantially higher than the next highest leverage value of approximately 0.0202. This is the same leverage value as in Question 3 because leverage depends on the X values, which have not changed.

6. Here, X = 500 is a high leverage point and an influential point. Why?

Answer The observation with X = 500 has high leverage because its X value is far outside the range of the other observations. It is also highly influential because its Y value of −600 is far from the original regression relationship. Adding this point substantially changes the regression coefficients, reversing the slope from 0.4932 to −0.8426.

7. What has caused the slope of the regression line to change? (Explain this as intuitively as you can.)

Answer The slope changed because the new observation has an extremely large X value but a very low Y value, which goes against the positive relationship in the original data. To reduce the large squared residual from this point, the OLS regression line is pulled downward on the right side, changing the slope from positive to negative.

Question 5

Add two new data points to the original dataset, with (X = 50, Y = 300) and (X = 50, Y = -200). They are both outliers. Dropping either point primarily changes the parameter estimates. Dropping both points doesn't notably change the parameter estimates but increases R-squared. The 2 data points are influential, both individually and in combination. Their impact is to cause instability in the parameter estimates and in the overall model fit.

Create 100 bootstrap samples, by sampling with replacement from the 102 data points. For each sample, calculate R-squared and save the result. Summarize this set of 100 R-squared statistics with a histogram. How stable are the results to small changes in the input dataset? This type of stability is one element of robustness.

```
# Original 100 observations
data5 <- data1 %>%
  select(X, Y)

# Add two outliers
new_points <- data.frame(
  X = c(50, 50),
  Y = c(300, -200)
)

# Combine observations
data5 <- rbind(data5, new_points)
nrow(data5)
```

```
## [1] 102
```

```
tail(data5, 2)
```

```
##      X    Y
## 101 50  300
## 102 50 -200
```

```
set.seed(1)

# store 100 bootstrap R-squared values
r2_boot <-  numeric(100)


# Generate 100 bootstrap samples
for (i in 1:100){
  boot_index <- sample(1:nrow(data5),
                       size = nrow(data5),
                       replace = TRUE)
  boot_data <- data5[boot_index,]

  #fit regression model
  boot_fit <- lm(Y ~ X, data = boot_data)

  # R-squared
  r2_boot[i] <- summary(boot_fit)$r.squared
```

```
}

head(r2_boot, 10)
```

```
##  [1] 0.12608391 0.06994965 0.12267156 0.04436214 0.10042817 0.03817558
##  [7] 0.27574902 0.27406446 0.60019538 0.51440624
```

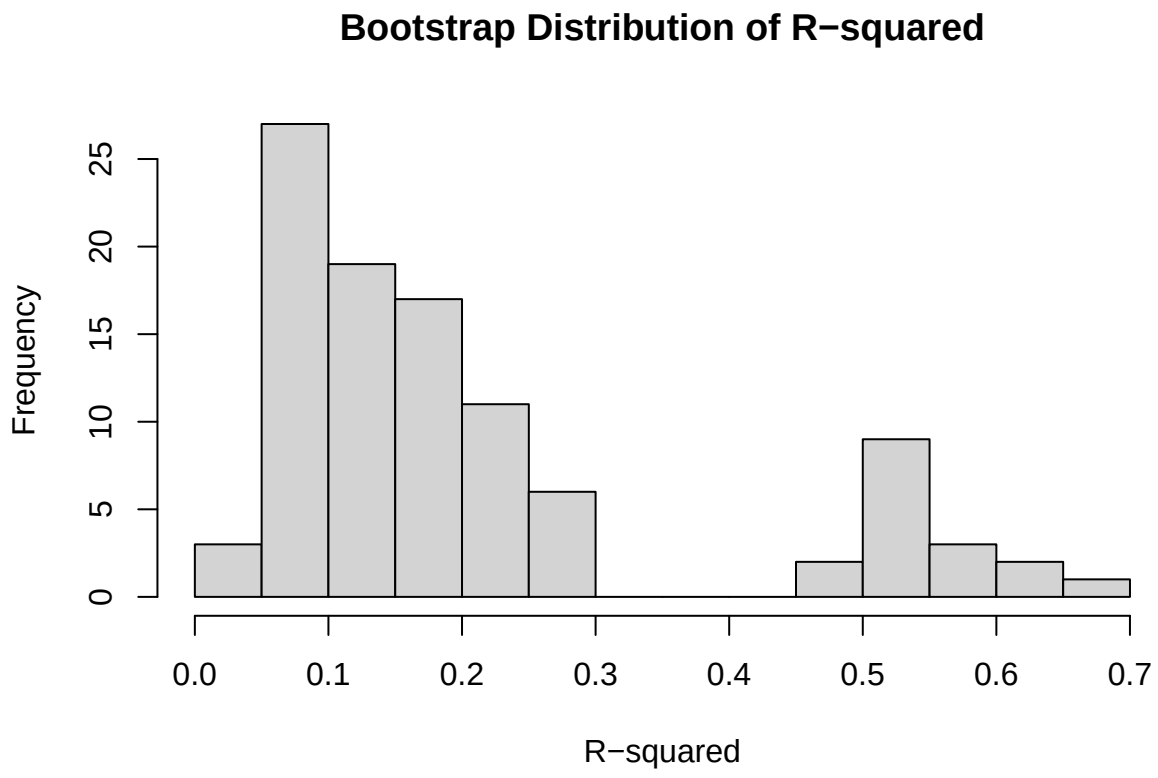
```
summary(r2_boot)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.02987 0.09193 0.15626 0.20888 0.23697 0.67972
```

```
sd(r2_boot)
```

```
## [1] 0.1690513
```

```
# Histogram of 100 bootstrap R-squared values
hist(r2_boot,
     main = "Bootstrap Distribution of R-squared",
     xlab = "R-squared",
     ylab = "Frequency",
     breaks = 15)
```



Answer The bootstrap results suggest that the model is not very stable. Across 100 bootstrap samples, R-squared ranged from approximately 0.030 to 0.680, with a mean of 0.209 and a standard deviation of 0.169. The histogram shows substantial variation, with most values concentrated between 0.05 and 0.30 and a smaller cluster around 0.45 to 0.70. This suggests that the model fit is sensitive to changes in the sample composition. The varying representation of the two outliers across bootstrap samples likely contributes to this instability, indicating limited robustness.